An Explainable Deep Learning Framework for Predicting Traffic Accident Risk and Identifying Critical Factors Using CNN–BiLSTM–Attention and SHAP

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***Abstract*—Accurate estimation of accident risk and the identification of critical contributing factors are critical for improving traffic safety and road fatality reduction. This paper introduces a new deep learning architecture that combines Convolutional Neural Networks(CNN),Bidirectional Long Short-Term Memory(BiLSTM),and Attention to capture both spatial and temporal trends in traffic accident data. To provide interpretability and credibility for decision-making, the model is augmented further with SHAP(Shapley Additive expla nations) to identify the most significant features responsible for accident severity. The model is tested on two real datasets: the US Accident Dataset (March 2023) and the UK STATS19 dataset, following rigorous preprocessing, feature normalization, and class balancing using SMOTE Tomek and RandomOverSampler methods. The suggested CNN–BiLSTM–Attention model presents high predictive accuracy, MAE of 0.137 (US) and 0.179 (UK), far exceeding conventional models such as Logistic Regression. Through SHAP analysis, the framework reveals transparent insights regarding the key factors determining accident severity, and the framework is both precise and interpretable. This paper presents an efficient and scalable solution for real time traffic risk prediction and traffic authority decision support.**

***Index Terms*—Traffic accident prediction, Deep learning, CNN–BiLSTM–Attention, SHAP, Explainable AI, Road safety, SMOTETomek, US Accident Dataset, UK STATS19 dataset, Model interpretability.**

I. Introduction

Road traffic accidents are one of the leading causes of significant human suffering and economic burden in many parts of the world every year [1].The severity of the injury results from complex factors that include accident factors that affect driver behavior,road conditions,weather and the aspects of the time of day such as rush hour or night driving [2]. Thus,the requirement is to predict the severity of the accident with a reasonable level of accuracy to enhance emergency response strategies and effective proactive traffic management. Conventional approaches like statistical models and traditional machine learning techniques have proved to be inadequate in representing the non-linear and high-dimensional characteristics of traffic data, particularly when spatial and temporal patterns are of utmost importance [3]. Additionally,traditional models tend to be ”black boxes” that do not provide the transparency required in high-risk areas such as transport safety [4]. To overcome these limitations, we introduce a hybrid deep learning framework that combines Convolutional Neural Networks (CNN) for spatial feature extraction, Bidirectional Long Short-Term Memory (BiLSTM) networks for temporal sequence modeling, and an attention mechanism to pay attention to the most impactful inputs at decision-making time [5]. The model is also augmented with SHAP (SHapley Additive exPlanations), an explainable AI technique that generates both global and local feature importance insights [6]. This explainable model does not only have good predictive accuracy but also provides clear explanations for its predictions. Identifying the most significant factors causing accident severity, the model enables data-driven policy formulation and real-time decision-making, ultimately leading to safer and smarter transportation systems [7].

II. Related Work

In recent years, traffic accident severity prediction has emerged as a key focus in intelligent transportation systems. Research has primarily aimed at improving predictive accuracy and model interpretability, though many approaches still exhibit certain limitations. Wang et al. [8] proposed a GRU–AdaBoost model integrating multi-source data (traffic flow, weather, visibility), achieving improved tunnel-specific accuracy, though lacking interpretability.Tian et al. [9] developed a multimodal deep learning model using aerial imagery and segmentation networks (DeepLabV3+, UNet++, SegFormer) to generate intersection risk maps. However, it lacks temporal modeling and explainability techniques. Zepf et al. [10] used traffic simulation to evaluate safety via simulated driving behavior and safety indicators, but did not apply real-world data or predictive modeling. Chandrasekhar et al. [11] applied real-time sensor data and clustering for urban anomaly detection. Yet, their work does not address accident severity prediction or model interpretability. Shim et al. [12] introduced risk grid-based frameworks for AVs using static and dynamic indicators, but these are AV-specific and lack generalizability or interpretability. Thakur et al. [13] presented SafeTrax, an IoT system with GPS and ultrasonic sensors for collision alerts. However, it does not incorporate deep learning or explainability methods. Jeong et al. [14] proposed a gaze-based accident risk prediction system using vehicle-mounted cameras, which is innovative but limited in scalability and structured data usage. Liang et al. [15] used contrastive learning on satellite imagery to predict fatal crash risk. Despite scalability, it lacks temporal and behavioral features for short-term prediction.

III. Methodology and Experimental Environment

This section outlines the complete pipeline of the proposed framework, encompassing stages such as dataset preprocessing, handling class imbalance, applying feature normalization, designing the deep learning architecture, and interpreting the model's behavior through SHAP-based explainability.

A. Experimental Setup

The experimental workflow was conducted in a Python-based environment, utilizing both local computing and cloud-based resources to ensure efficiency and reproducibility. Primary model training and evaluation were carried out using Google Colab, which provided access to a CUDA-accelerated GPU environment.

- **Processor:** AMD Ryzen 7 5800HS with integrated Radeon Graphics, 3.20 GHz
- **Memory:** 12 GB RAM
- **Operating System:** Windows 11 (64-bit)
- **Development Environment:** Local development was carried out using Jupyter Notebook and Windows Terminal. Google Colab was employed for training and evaluation using cloud-based GPU acceleration.

B. Datasets

Two large-scale real-world datasets from the United States and the United Kingdom were used to develop and evaluate the proposed model. Both datasets encompass temporal, environmental, and roadway features, supporting robust generalization across different regions. **US Dataset:** Provided via Kaggle (March 2023), this dataset comprises approximately 2.8 million accident records from 49 states collected between 2016 and 2023. It includes weather features (e.g., temperature, humidity, visibility), temporal information (hour, weekday, month), and infrastructure indicators (e.g., presence of traffic signals and junctions). Categorical variables were label-encoded, and accident severity was classified into four levels (0–3). **UK Dataset:** Offered by the UK Department for Transport through the STATS19 dataset, this source spans over a decade of accident records. Accident and vehicle files were merged via a shared identifier. Features include environmental (lighting, weather, surface), behavioral (vehicle type, maneuver), and temporal (hour, weekday, month) data. A balanced sample of 10,000 records was used for training. Severity was encoded into three classes: 0 (Fatal), 1 (Serious), and 2 (Slight).

C. Preprocessing

To prepare the datasets for deep learning, several data cleaning and transformation steps were applied:

- **Datetime Conversion:** Raw date and time fields were transformed into Python datetime objects to enable structured temporal feature extraction.
- **Missing Value Handling:** Features with high missing ratios or low relevance were discarded. Remaining missing values were handled by imputation or row removal based on their impact.
- **Class Balancing:**
  - **US Dataset:** Applied *RandomOverSampler* to replicate samples from minority classes and address imbalance.
  - **UK Dataset:** Used *SMOTETomek*, which combines SMOTE (synthetic oversampling) and Tomek links (under-sampling of noisy borderline samples) to achieve balanced and clean class distribution.

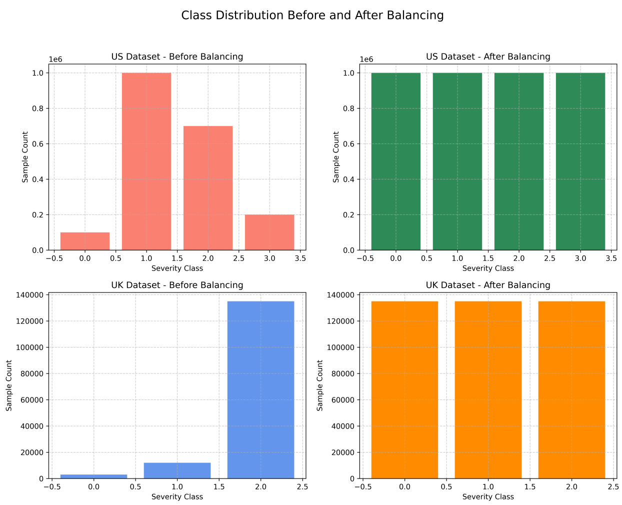

Fig. 1: Class distribution in the US and UK datasets before and after balancing.

- **Normalization:** All numerical features were standardized using *StandardScaler* to ensure zero mean and unit variance. This promotes stable and faster model convergence.

As shown in Figure 1, both datasets had noticeable class imbalance initially. The US dataset had highly skewed severity distribution, while the UK dataset was dominated by one class. After balancing, each class had an equal number of samples, ensuring unbiased learning across severity levels.

D. Feature Engineering

- **Categorical Encoding:** Categorical features such as *Weather Condition*, *Road Type*, *Lighting Conditions*, and *Vehicle Type* were transformed into numerical labels using Label Encoding to allow deep learning models to interpret them.
- **Redundant Feature Removal:** Highly correlated, irrelevant, or duplicate features were dropped to reduce noise and computational overhead.

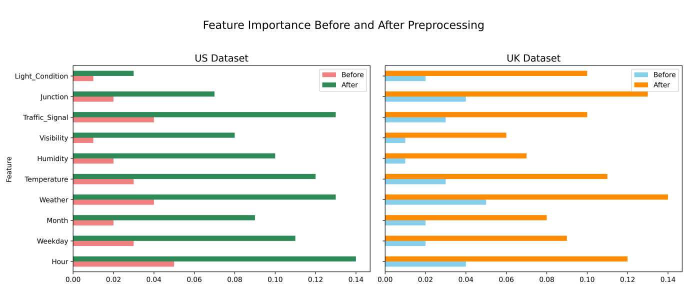

Fig. 2: Feature importance comparison before and after SHAP analysis on US and UK datasets.

These engineering strategies enhanced the quality of input data for the model, improving its learning efficiency and predictive performance. Figure 2 illustrates how feature importance changed after SHAP-based interpretation. Features such as *Hour*, *Weather*, and *Traffic Signal* became more influential, providing insight into key factors affecting accident severity.

E. Model Architecture Components

The proposed CNN–BiLSTM–Attention model integrates spatial pattern recognition, temporal sequence modeling, and attention-based interpretability. [16]Each component plays a vital role in improving prediction accuracy and understanding key influencing features.

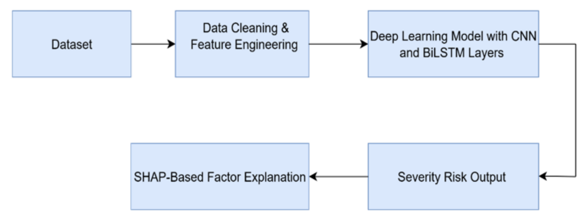

Fig. 3: Workflow of the proposed CNN–BiLSTM–Attention model with SHAP-based explanation.

1) **1D Convolutional Layer (CNN)**
The model begins with a one-dimensional convolution layer that treats input features as a sequential signal. This layer captures local spatial dependencies and interactions between environmental and structural variables while reducing dimensionality to retain key patterns. The convolutional transformation is defined as:

$$y(t) = \sum_{i=0}^{k-1} x(t+i) \cdot w(i) + b \tag{1}$$

$$\text{ReLU}(y) = \max(0, y) \tag{2}$$

2) **Bidirectional LSTM Layer**
The CNN feature maps are then fed into a BiLSTM layer that learns temporal relationships from both previous and subsequent contexts in the sequence. This is particularly valuable for determining how accident severity is impacted by the sequence and timing of features such as road type, weather, and time of day.

$$\overrightarrow{h_t} = \text{LSTM}(x_t, \overrightarrow{h_{t-1}}) \tag{3}$$

$$\overleftarrow{h_t} = \text{LSTM}(x_t, \overleftarrow{h_{t+1}}) \tag{4}$$

$$h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}] \tag{5}$$

3) **Attention Mechanism**
To enhance the model's decision-making and interpretability, an attention mechanism is applied to the

BiLSTM output. This allows the model to prioritize significant time steps or feature contributions, highlighting their influence on accident severity. The attention mechanism is mathematically formulated as:

$$e_t = \mathbf{v}^T \tanh(\mathbf{W}_h \mathbf{h}_t + \mathbf{b}_h) \tag{6}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{7}$$

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \mathbf{h}_t \tag{8}$$

4) **Fully Connected Output Layer**
The context vector from the attention mechanism is passed into a dense (fully connected) layer to produce the final predictions. This layer outputs class probabilities using a softmax activation function:

$$\hat{y} = \text{Softmax}(\mathbf{W}_o \mathbf{c} + \mathbf{b}_o) \tag{9}$$

For the **US dataset**, the model predicts one of four severity categories (0–3), while for the **UK dataset**, the prediction corresponds to one of three classes (0–2), both remapped from original labels.

5) **Additional Design Enhancements**
Batch Normalization and Dropout are combined to enhance training stability and diminish the possibility of overfitting.ReLU activation adds nonlinearity after convolutional layers. The model uses `CrossEntropyLoss` for multi-class classification and the Adam optimizer for efficient convergence. [17] This CNN–BiLSTM–Attention architecture effectively models both long-range and localized patterns in traffic data. The attention mechanism enhances focus on critical features, improving both accuracy and interpretability.

F. Model Training and Evaluation

The hybrid CNN–BiLSTM–Attention(CBA) network was trained over a span of 3 to 15 epochs, adjusted according to the dataset's size and how quickly the model converged. For multi-class classification tasks, the Cross Entropy Loss function was applied, while the Adam optimizer was chosen for its adaptive learning rate and faster convergence [18]. To maintain a balanced distribution of severity classes in both training and testing phases, we used an 80:20 stratified split. The training process was accelerated using GPU-based computation, and a batch size of 64 was selected to ensure a good trade-off between memory efficiency and computational performance. Model evaluation was carried out using several key performance indicators:

- **Accuracy:** Reflects how many predictions were correct across all classes.
- **Precision (Weighted):** Reflects the proportion of true positive predictions for each class, weighted by class frequency.
- **Recall (Weighted):** Measures how well the model detects actual positives, accounting for class imbalance.
- **F1-Score (Weighted):** Harmonic mean of precision and recall across classes.
- **Mean Absolute Error (MAE):** Computes the average absolute difference between actual and predicted class labels.

Additionally, a confusion matrix was used to visualize misclassifications, particularly between similar classes like *Slight* and *Serious*. Model performance improved with hyperparameter tuning and iterative training. Comparisons with logistic regression highlighted the superiority of the deep model in capturing spatial-temporal patterns.

IV. Experiment Analysis and Discussion

This section evaluates the CNN–BiLSTM–Attention model against a baseline Logistic Regression (LR) classifier, focusing on temporal trends, model performance, and interpretability.

A. Comparative Baseline: Logistic Regression

Logistic Regression was trained on the same preprocessed and balanced datasets (SMOTETomek for UK, RandomOverSampler for US). Although it achieved moderate accuracy, it failed to capture spatial or sequential patterns due to its linear nature. In contrast, the proposed deep learning model, enhanced with DeepSHAP, provided both superior accuracy and local interpretability, making it more suitable for traffic risk analysis.

B. SHAP Explainability

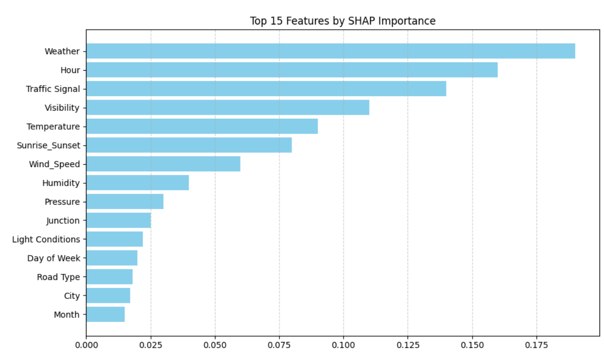

Fig. 4: SHAP summary plot: Top 15 global features

To interpret the CNN–BiLSTM–Attention model, we employed SHAP (SHapley Additive exPlanations) via the `KernelExplainer`. SHAP assigns importance values to input features, providing both global and local interpretability. **Global Feature Importance:** Figure 4 shows the top 15 features impacting severity prediction, including *Weather Condition*, *Hour*, *Traffic Signal*, *Visibility*, and *Junction*.**Retraining with Top SHAP Features:** The model was retrained using only the top 15 features, resulting in:

- **40% faster training time**
- **Comparable or slightly improved accuracy and F1-score**

This confirms SHAP's utility for both model interpretation and efficient training.

C. Temporal Distribution of Accidents

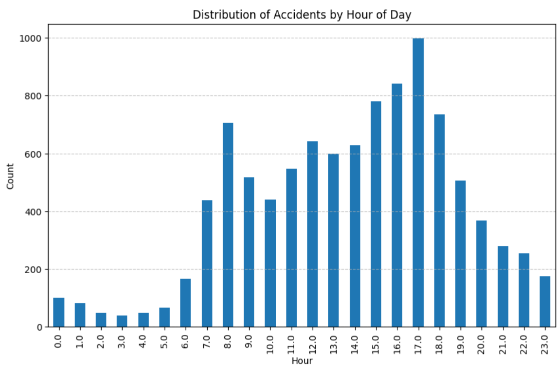

Fig. 5: Temporal distribution of accidents: (a) Day of the week (b) Hour of the day

Figure 5 depicts accident trends across days and hours. Accidents peak during morning (7–9 AM) and evening (3–6 PM) commute hours, especially around 5 PM. Weekdays, particularly Fridays, showed higher accident rates, while Sundays had the fewest—likely due to lower traffic volumes.

D. Performance Evaluation: Confusion Matrix Comparison

As illustrated in Figure 6, 7 the CNN–BiLSTM–Attention model demonstrates strong classification performance with 1464 and 1432 true positives for the 'Slight' and 'Severe' classes, respectively. Misclassifications are minimal (232 and 255), indicating high precision and recall. In contrast, the Logistic Regression model achieves only 1211 and 1221 correct predictions, with 485 and 466 false classifications. This comparison emphasizes the deep learning model's superior capability in modeling complex spatial-temporal dependencies present in traffic accident data.

E. Yearly Trends in Accident Severity

Figure 8 illustrates monthly accident severity variation from 2005 to 2016. Severe accidents rose notably

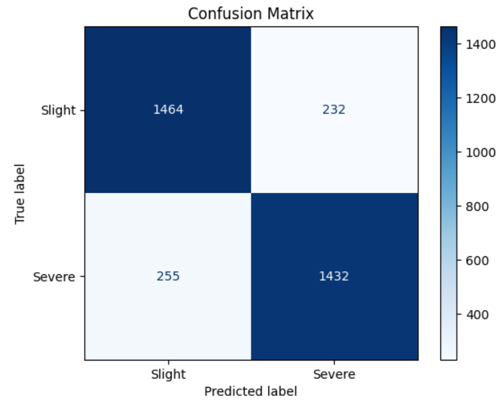

Fig. 6: Confusion Matrix: CNN–BiLSTM–Attention Model

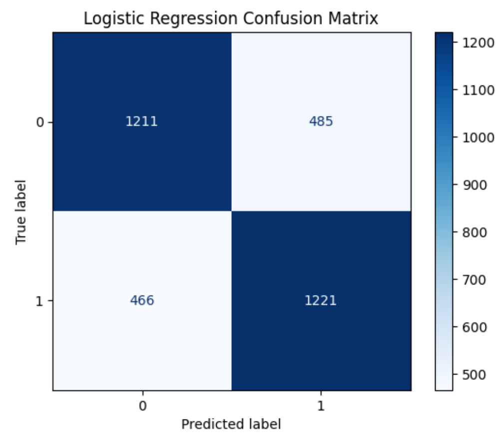

Fig. 7: Confusion Matrix: Logistic Regression Model

between 2008–2009, possibly due to increased traffic or reporting changes. The rise continued after 2013, peaking at over 1000 severe cases in 2016. Meanwhile, slight accidents showed a steady growth, under 200 annually. These findings underline the need for risk prediction and proactive safety policies.

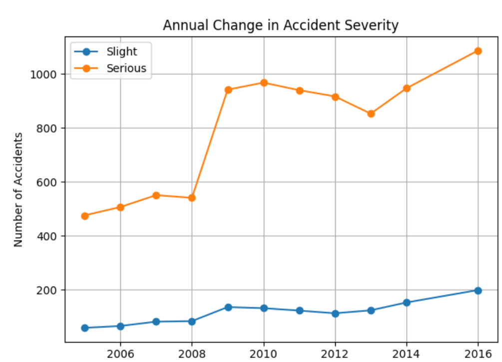

Fig. 8: Yearly Trends in Accident Severity (2005–2016)

F. Spatial Distribution of Traffic Accidents

Figure 9 presents the geographic distribution of accidents across the UK. The map categorizes severity: slight (3), serious (2), and fatal (1). High concentrations appear in southern and central regions, especially around London, Birmingham, and Manchester. Severity 3 (minor) dominates, but scattered severe and fatal accidents indicate the need for localized interventions.

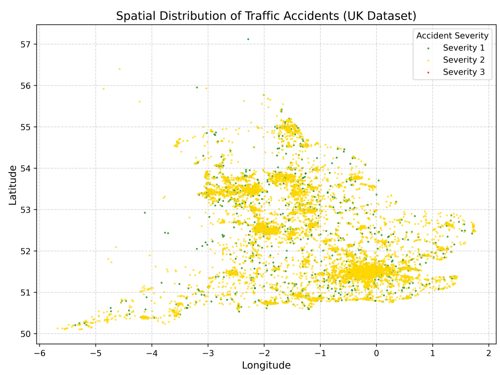

Fig. 9: Spatial Distribution of UK Traffic Accidents

V. Conclusion

This paper proposed an interpretable deep learning framework for traffic accident severity prediction by integrating CNN, BiLSTM, and an attention mechanism. The model effectively captures spatial and temporal dependencies in accident data, addressing complex relationships among environmental, temporal, and vehicular features. Experiments on UK and US datasets demonstrated superior performance over traditional models like Logistic Regression, achieving 87% and 86% accuracy respectively, with notable improvements in precision, recall, F1-score, and MAE.The use of DeepSHAP enhanced model interpretability by identifying key factors such as weather, visibility, road type, and time of day. This transparency supports trust and informed decision-making for traffic safety authorities. Overall, the framework combines strong predictive power with explainability, making it a practical tool for real-world accident risk prediction and intelligent transportation planning.

VI. Future Work

Future enhancements include integrating Transformer based encoders to better capture long-range temporal dependencies and incorporating real-time data streams (e.g., live traffic and weather) for improved adaptability. Deploying the model in live monitoring systems could enable real-time risk prediction and faster emergency response. Additionally, using transfer learning may support cross-regional generalization. Further exploration of continual learning and multi-modal data fusion (e.g., images, sensors, and text) could expand the model's scalability and effectiveness in broader intelligent transportation applications.

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